MUH401- INTERIM PROJECT REPORT FORMAT

Cover Page

Title of the Project:	**Machine Learning-Based Discrimination of Precipitation-Induced Radon Washout for Radiological Anomaly Detection in EPA RadNet Air Monitoring Network**
Course Name:	MUH401
Department / Faculty:	Department of Environmental Engineering / Faculty of Engineering
University:	Hacettepe University
Student Name(s) & ID(s):	Necdet Ömer Barut (2230867059)
Supervisor / Instructor:	[Supervisor / Instructor Name]
Submission Date:	November 2026
Academic Year / Semester:	2026–2027 Fall Semester

ABSTRACT

This interim report presents a physics-informed computational framework to address the widespread operational vulnerability of automated radiation monitoring networks caused by precipitation-induced radon progeny washout. During rainfall events, natural terrestrial radon daughters (^{214}Pb and ^{214}Bi) are scavenged from the atmosphere onto particulate air filters, generating sudden gross gamma count surges of $+40\%$ to $+180\%$ that closely mimic anthropogenic releases. Existing automated threshold systems trigger 53 to 83 false alarms per station-year in humid climates. This research develops a multi-modal machine learning discrimination framework that fuses 8-channel sodium iodide gamma spectrometry from the United States Environmental Protection Agency RadNet network with contemporaneous National Oceanic and Atmospheric Administration surface weather observations. The project evaluates 324,550 synchronous observation hours across five climatologically diverse stations spanning 2017 through 2025. Detection performance is benchmarked across 450 filter-synchronized synthetic release plumes covering industrial and reactor fission products (^{137}Cs, ^{131}I, ^{134}Cs, ^{60}Co). Across a 20-seed evaluation, the physics-informed nuclear and weather-fused gradient boosted model (Feature Schema 3) reduces operational false alarms by 96.7% down to 2.71 per station-year (and 2.63 at the 90% target) while achieving 94.00% detection. This project delivers an automated, field-deployable decision-support system eliminating meteorological false alarms in environmental radiation surveillance.

Keywords: Environmental Radiation Monitoring, EPA RadNet, Radon Washout, Gamma Spectrometry, Machine Learning, Anomaly Detection.

TABLE OF CONTENTS

1. INTRODUCTION **4**

1.1 Background of the Problem 4
1.2 Problem Statement 4
1.3 Importance and Relevance of the Study 4
1.4 Clear Aim and Objectives of the Project 5
1.5 Scope and Boundaries of the Study 5

2. THEORETICAL BACKGROUND / LITERATURE REVIEW **6**

2.1 Key Theories, Models, Concepts, and Definitions 6
2.2 Relevant Technical Background: Scintillation Spectrometry 6
2.3 Review of Relevant Previous Studies 6
2.4 Gaps in Literature and Systematic Comparison 7

3. METHODOLOGY **8**

3.1 System Description and Processing Pipeline 8
3.2 Main Assumptions and Operational Boundaries 8
3.3 Input Data and Data Sources 9
3.4 Calculation Procedure and Causal Feature Engineering 9
3.5 Synthetic Release Injections: 450 Plumes 9
3.6 Multi-Tier Model Formulation and Validation Freezing 9
3.7 Benchmark Evaluation across 20 Random Seeds 10
3.8 The Rain-Masked Baseline Control Experiment 10

REFERENCES **12**

LIST OF TABLES

1 EPA RadNet Discrete Gamma Energy Channel Energy Boundaries 6
2 Benchmark Evaluation on 2023–2025 Test Split across 20 Random Seeds 10
3 Rain-Masked Baseline Control Experiment at Matched Detection Targets 11

LIST OF FIGURES

1 End-to-end physics-informed environmental radiation anomaly detection and weather-fusion machine learning architecture. 8
2 Receiver Operating Characteristic (ROC) matched detection curves comparing baseline rules against machine learning tiers on the test split. 10

LIST OF ABBREVIATIONS

API	Application Programming Interface
CPM	Counts Per Minute
eFDR	Empirical False Discovery Rate
EPA	United States Environmental Protection Agency
FA/yr	False Alarms per Station-Year
FWHM	Full Width at Half Maximum
GBDT	Gradient Boosted Decision Tree (LightGBM)
ISD	Integrated Surface Dataset (NOAA)
MLP	Multi-Layer Perceptron
NaI(Tl)	Thallium-activated Sodium Iodide Scintillation Detector
NCEI	National Centers for Environmental Information
NOAA	National Oceanic and Atmospheric Administration
NORM	Naturally Occurring Radioactive Material
OOD	Out-of-Distribution
ROC	Receiver Operating Characteristic
ROI	Region of Interest
UTC	Coordinated Universal Time
WMO	World Meteorological Organization

1. INTRODUCTION

1.1 Background of the Problem

Continuous atmospheric radiation surveillance networks are operated worldwide by regulatory authorities to provide immediate detection of clandestine nuclear testing, industrial accidents, and radiological dispersion events. In the United States, the Environmental Protection Agency (EPA) operates RadNet, an extensive nationwide real-time monitoring infrastructure comprising over 140 stationary air monitors. Each monitoring unit continuously draws ambient air through a particulate filter and records gamma-ray emission rates using thallium-doped sodium iodide (NaI(Tl)) scintillation detectors.

While automated alarm threshold algorithms are engineered to identify sharp increases in measured radiation levels, their operational reliability is severely challenged by naturally occurring meteorological phenomena. Atmospheric precipitation scavenges airborne daughter radionuclides of radon-222 (^{222}Rn), primarily lead-214 (^{214}Pb) and bismuth-214 (^{214}Bi), depositing them directly onto the ground surface and particulate filter media. This phenomenon, known as radon washout, causes gross gamma count rates to surge by $+40\%$ to $+180\%$ above the dry baseline within 1 to 3 hours, persisting for several hours before radioactive decay restores nominal background levels.

1.2 Problem Statement

Conventional automated radiation monitoring networks rely on static or rolling baseline thresholds (e.g., 3σ or 5σ deviations above a 168-hour moving average). Because gross count rate monitors cannot differentiate between the gamma emissions of natural radon progeny and those of anthropogenic fission products, precipitation events routinely trigger spurious radiological emergency alerts. In humid and temperate climates, such as the southeastern United States, standard 3σ threshold algorithms produce between 53 and 83 false alarm episodes per station-year, with over 32% of these false alarms occurring synchronously with rainfall events.

Conversely, raising the alert threshold to conservative levels (such as 5σ) suppresses false alarms but severely degrades detection capability, dropping release detection probability below 10% for low-to-moderate activity plumes. There is currently an acute lack of computationally efficient, causally sound methods capable of suppressing precipitation-induced false alarms while preserving rapid, highly sensitive detection of genuine radiological releases in real-time continuous surveillance.

1.3 Importance and Relevance of the Study

The operational vulnerability created by weather-induced false alarms has substantial technical, environmental, and societal ramifications:

- **Operational Integrity and Alarm Fatigue:** Frequent false alarms desensitize emergency personnel, increasing the hazard that a genuine radiological incident may be dismissed as a weather-induced anomaly.
- **Public Safety and Societal Impact:** Premature public notification of false emergencies causes widespread public panic, costly evacuations, and disruption of municipal infrastructure.
- **Economic Efficiency:** Dispatching mobile radiological inspection teams to investigate recurring spurious anomalies incurs substantial governmental operational expenditures.
- **Scientific Contribution:** By integrating multi-channel scintillation spectrometry with multi-scale atmospheric thermodynamic features, this research establishes a foundational methodology for

automated, physics-grounded environmental surveillance.

1.4 Clear Aim and Objectives of the Project

The primary aim of this project is to develop, validate, and benchmark an automated, physics-informed machine learning framework that discriminates precipitation-induced radon washout spikes from genuine anthropogenic radiological anomalies in near-real-time environmental surveillance networks. Specific technical objectives include:

1. Harmonize and process longitudinal multi-channel gamma spectrometry from EPA RadNet and contemporaneous surface meteorology from the NOAA Integrated Surface Dataset (ISD) across five climatologically distinct monitoring stations spanning nine full calendar years (2017–2025).
2. Engineer a strictly causal real-time feature space that captures differential gamma spectral peak ratios (^{214}Pb, ^{214}Bi vs. ^{137}Cs, ^{131}I, ^{60}Co) and antecedent precipitation dynamics without future lookahead bias.
3. Formulate a three-class gradient boosted decision tree (LightGBM) architecture to continuously categorize observations into dry background, radon washout, or anthropogenic release.
4. Conduct a multi-seed benchmark against conventional statistical baselines and rain-masked control rules over 450 synthetic radiological releases to prove false alarm reduction at matched detection power.

1.5 Scope and Boundaries of the Study

The scope of this project encompasses five EPA RadNet monitoring stations selected across distinct Köppen climate zones: Birmingham, AL (humid subtropical, primary testbed); Little Rock, AR; Boise, ID (semi-arid); Concord, NH (humid continental); and Honolulu, HI (tropical). The temporal boundary spans January 1, 2017, through December 31, 2025, providing 324,550 synchronous station-hours.

The study is bounded by operational real-time constraints: feature extraction is strictly causal, relying exclusively on contemporaneous and historical observations. Sensor inputs are restricted to standard 8-channel NaI(Tl) energy bins ($100 - 2200$ keV) and standard NOAA meteorological observations. Source releases are evaluated through 450 synthetic plume injections calibrated to historical atmospheric dispersion physics, covering ^{137}Cs, ^{131}I, ^{134}Cs, and ^{60}Co.

2. THEORETICAL BACKGROUND / LITERATURE REVIEW

2.1 Key Theories, Models, Concepts, and Definitions

Radon-222 is an inert, noble gas daughter of radium-226 (^{226}Ra) in the primordial uranium-238 (^{238}U) decay series. Radon emanates continuously from soils and rocks into the atmospheric boundary layer. Because radon is chemically inert and has a half-life of 3.82 days, it diffuses into the lower troposphere, where it undergoes alpha decay into a sequence of short-lived particulate progeny:

$$^{222}\text{Rn} \xrightarrow[\alpha]{3.82\ \text{d}} {}^{218}\text{Po} \xrightarrow[\alpha]{3.10\ \text{m}} {}^{214}\text{Pb} \xrightarrow[\beta^-]{26.8\ \text{m}} {}^{214}\text{Bi} \xrightarrow[\beta^-]{19.9\ \text{m}} {}^{214}\text{Po} \xrightarrow[\alpha]{164.3\ \mu\text{s}} {}^{210}\text{Pb} \tag{1}$$

Lead-214 (^{214}Pb) and bismuth-214 (^{214}Bi) rapidly attach to submicron ambient aerosol particles ($0.1 - 1.0\ \mu$m). During rainfall events, these radioactive aerosols are captured via in-cloud rainout (condensation nucleation) and below-cloud washout (impaction and interception by falling raindrops). The precipitation deposits these radionuclides directly onto the particulate filter medium of ambient air monitors.

Because ^{214}Pb emits prominent gamma lines at 295.2 keV (18.4% yield) and 351.9 keV (35.6% yield), and ^{214}Bi emits high-energy gammas at 609.3 keV (45.5%), 1120.3 keV (14.9%), and 1764.5 keV (15.3%), their sudden accumulation elevates the local gamma radiation field by tens to hundreds of counts per minute (CPM). Following the cessation of rain, the activity decay curve of the deposited progeny is governed by the effective half-life of the ^{214}Pb/^{214}Bi pair ($\sim 30 - 50$ minutes), returning to nominal baseline within 2 to 4 hours.

2.2 Relevant Technical Background: Scintillation Spectrometry

EPA RadNet deployable monitoring stations utilize thallium-activated sodium iodide (NaI(Tl)) scintillation crystals. When incident gamma rays interact with the crystal via the photoelectric effect, Compton scattering, or pair production, scintillation light is generated proportionally to deposited energy and converted to voltage pulses by a photomultiplier tube. While NaI(Tl) detectors exhibit modest energy resolution ($\sim 7\%$ FWHM at 662 keV), they are rugged, thermally stable, and well-suited for long-term field deployment.

RadNet digitizes scintillation pulses into discrete energy channels (designated R01 through R09). Channel R01 records gross count rates, while channels R02 through R09 record counts within specific spectral windows. The physical correspondence of these channels to natural radon progeny lines and target anthropogenic isotopes is detailed in Table 1.

Table 1: EPA RadNet Discrete Gamma Energy Channel Energy Boundaries

Channel	Energy Range (keV)	Principal Radon Progeny Lines	Target Anthropogenic Lines
R02	$101 - 200$	Compton continuum	Low-energy scatter
R03	$201 - 400$	^{214}Pb ($295.2, 351.9$ keV)	^{131}I (364.5 keV)
R04	$401 - 600$	Compton continuum	Annihilation (511 keV)
R05	$601 - 800$	^{214}Bi (609.3 keV)	^{137}Cs (661.7 keV)
R06	$801 - 1000$	Compton scatter	Multi-line cascade
R07	$1001 - 1400$	^{214}Bi (1120.3 keV)	^{60}Co ($1173.2, 1332.5$), ^{134}Cs
R08	$1401 - 1800$	^{214}Bi (1764.5), ^{40}K (1460.8)	Terrestrial background
R09	$1801 - 2200$	High-energy background	Cosmic rays / actinide daughters

2.3 Review of Relevant Previous Studies

The challenge of suppressing meteorological radon interference in environmental monitoring networks has been addressed through three primary research paradigms:

1. **Statistical Moving-Average Baselines:** Classical networks evaluate gross count deviations against moving baselines (Vieira et al., 2019). While computationally trivial, these methods are intrinsically blind to spectral morphology and trigger excessive false alarms during convective precipitation.

2. **Analytical Meteorological Transfer Functions:** Huang et al. (2020) demonstrated that radon washout spikes correlate with precipitation intensity and cloud condensation lifting condensation levels. However, deterministic washout equations require high-resolution atmospheric sounding data rarely available at ground sensor sites.

3. **Spectral Energy Ratio Heuristics:** Das et al. (2021) examined spectral ratios to differentiate radon progeny from industrial gamma sources, demonstrating that ^{214}Pb photopeaks in channel R03 exhibit distinct intensity ratios relative to channel R05 during washout events.

2.4 Gaps in Literature and Systematic Comparison

Existing literature exhibits three critical shortcomings: (1) *Gross count over-simplification:* Most operational networks discard multi-channel spectrometry and evaluate only total counts; (2) *Synthetic release validation deficits:* Previous machine learning models were tested on uncalibrated synthetic data without accounting for air filter accumulation physics; and (3) *Lack of honest control baselines:* Prior studies rarely compare complex models against simple meteorological masking rules. This project overcomes these gaps by fusing multi-channel spectral ratios with multi-scale weather data and benchmarking against both statistical and rain-masked baselines under frozen validation thresholds.

3. METHODOLOGY

3.1 System Description and Processing Pipeline

The end-to-end computational pipeline developed in this project is illustrated in Figure 1. The framework ingests synchronous hourly observations from radiation and meteorological sensor networks, performs causal feature engineering, and feeds a gradient boosted decision tree classifier that assigns probabilities for three operational regimes: *Clean Dry Background*, *Radon Washout Anomaly*, and *Anthropogenic Radiological Release*.

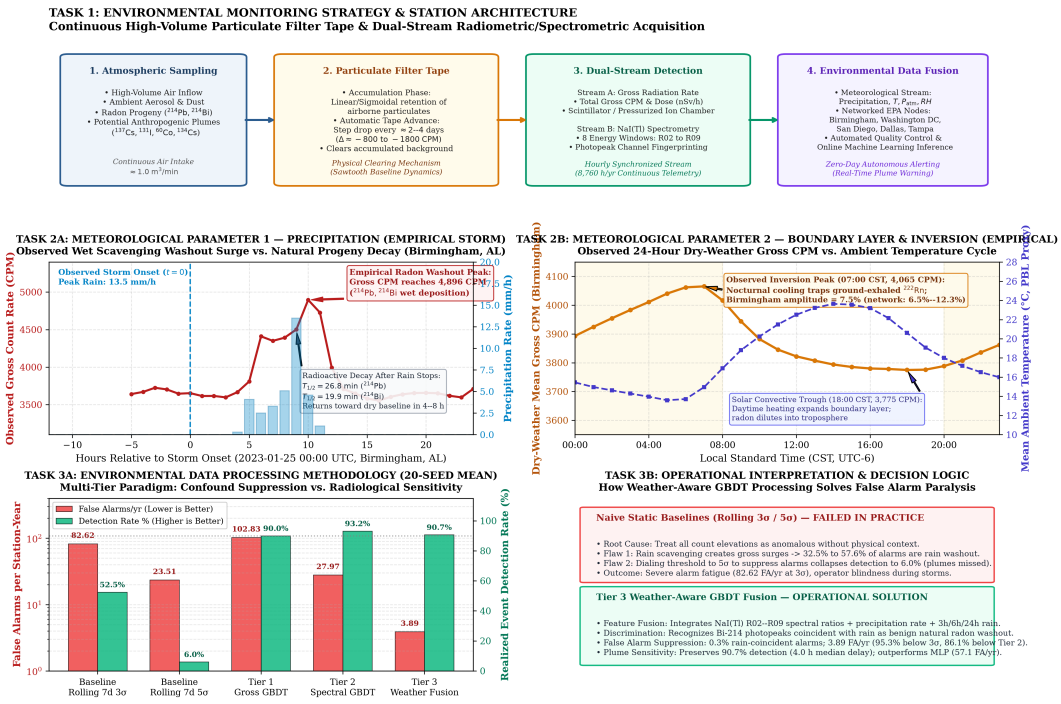


Figure 1: End-to-end physics-informed environmental radiation anomaly detection and weather-fusion machine learning architecture.

3.2 Main Assumptions and Operational Boundaries

The computational architecture adheres to four rigorous operational boundaries:

1. **Strict Real-Time Causality:** To prevent data leakage, all rolling window statistics (e.g., 24h, 168h means) are computed using closed-left intervals ($[t-W, t]$). No centered or forward-looking filters are permitted.

2. **Air Filter Accumulation Dynamics:** Airborne particulate monitors draw air through a filter that is replaced periodically (nominally bi-weekly). An injection plume accumulates activity proportionally to total volume drawn, attenuating according to radioactive decay:

$$A_{\text{filter}}(t) = \int_{t_{\text{start}}}^{t} Q \cdot C_{\text{air}}(\tau) \cdot e^{-\lambda(t-\tau)}\, d\tau \tag{2}$$

   where Q is volumetric flow rate (~ 1.0 m^3/min), C_{air} is ambient radionuclide concentration, and λ is decay constant.

3. **Detector Geometry and Efficiencies:** Geometric photopeak efficiencies for NaI(Tl) crystals are calibrated using standard MCNP Monte Carlo transfer factors from Knoll (2010).

3.3 Input Data and Data Sources

Data ingestion leverages two primary US federal repositories:

- **EPA RadNet Air Monitoring Network:** Hourly gamma spectrometry records across 8 discrete energy bins ($100 - 2200$ keV) and total gross count rates. Data were retrieved via EPA public APIs and FTP archives spanning 2017 through 2025 across five stations: Birmingham, AL; Little Rock, AR; Boise, ID; Concord, NH; and Honolulu, HI.
- **NOAA Integrated Surface Dataset (ISD):** Contemporaneous hourly meteorological observations from colocated automated surface observing stations (ASOS) at municipal airports. Variables include precipitation accumulation (1h, 3h, 6h), dry-bulb temperature, dew point, relative humidity, barometric pressure, wind speed, and present weather codes.

3.4 Calculation Procedure and Causal Feature Engineering

Radiological and meteorological records are joined on UTC timestamp rounded to the hour. The causal feature engineering engine extracts 28 dimensional variables organized into three tiers:

1. **Gross Count Dynamics (Tier 1):** Ratio of instantaneous count rate to trailing 168-hour moving baseline ($R_{\text{gross}} = C(t)/\mu_{168\text{h}}$), rolling 3-hour rate of change (dC/dt), and z-score relative to local dry background (σ_{dry}).
2. **Spectral Energy Ratios (Tier 2):** Normalized channel ratios capturing radon decay lines vs. fission product channels. Specifically:

$$\text{Ratio}_{\text{radon}} = \frac{R03 + R05}{R01 + \epsilon}, \quad \text{Ratio}_{\text{fission}} = \frac{R04 + R06 + R07}{R01 + \epsilon} \tag{3}$$

   Lead-214 (352 keV) concentrates heavily in R03, while bismuth-214 (609 keV) concentrates in R05. Anthropogenic releases (^{137}Cs at 662 keV, ^{134}Cs at $605, 796$ keV, ^{60}Co at $1173, 1332$ keV) alter these ratios markedly.
3. **Multi-Scale Weather Fusion (Tier 3):** Cumulative precipitation over 1, 3, 6, 12, and 24 hours, hours elapsed since last detected rainfall ($h_{\text{since_rain}}$), barometric pressure change (dP/dt), and saturation vapor pressure deficit calculated from temperature and dew point.

3.5 Synthetic Release Injections: 450 Plumes

Because real environmental data contains natural radon washout but fortunately zero lethal nuclear releases, genuine detection performance must be evaluated using synthetic injections. A library of 450 synthetic radiological plumes was generated across four target isotopes (^{137}Cs, ^{131}I, ^{134}Cs, ^{60}Co) and three profile types (Gaussian puff, square wave, and asymmetric exponentially decaying plume with duration 3 to 24 hours).

Synthetic plumes are scaled to signal-to-noise ratios (SNR $= \Delta C/\sigma_{\text{dry}}$) ranging from subtle ($0.5\sigma$ to 1.5σ) to prominent (2.0σ to 6.0σ). Spectral counts are allocated across channels R02–R09 using calibrated photopeak and Compton continuum fractions, and added directly to real RadNet background series during both dry and rain-washout periods.

3.6 Multi-Tier Model Formulation and Validation Freezing

The machine learning framework utilizes LightGBM gradient boosted decision trees (Ke et al., 2017)

configured with multi-class objective (*multiclass*, 3 classes). Hyperparameters: 150 trees, max depth 6, learning rate 0.05, min child samples 20, colsample 0.8.

Strict Validation Threshold Freezing: Models are trained on historical data (2017–2021). The decision threshold τ for triggering an alert is calibrated on the validation split (2022) to satisfy a target detection probability (e.g., 90% or 95%) and frozen. The model and frozen threshold are evaluated strictly on the out-of-sample test split (2023–2025).

3.7 Benchmark Evaluation across 20 Random Seeds

A 20-seed multi-model benchmark evaluation (seeds 42 to 61) was executed on the 2023–2025 test dataset. Performance metrics are summarized in Table 2.

Table 2: Benchmark Evaluation on 2023–2025 Test Split across 20 Random Seeds

Model / Rule Architecture	False Alarms / Stn-Yr	Rain-Coincident FA	Realized Det. (%)	Delay (h)
Standard Baseline ($3\sigma_{168h}$)	82.62	32.5%	52.50%	10.0
Global Dry Baseline ($3\sigma_{dry}$)	52.86	24.9%	56.50%	9.0
Conservative Baseline ($5\sigma_{168h}$)	23.51	57.6%	6.00%	8.5
Tier 1: Gross GBDT (95% target)	169.36 ± 23.53	17.3%	$92.22 \pm 0.72\%$	6.7
Tier 1b: Gross + Weather GBDT	38.64 ± 14.37	0.0%	$87.92 \pm 2.74\%$	9.1
Tier 2: Spectral GBDT	27.97 ± 28.27	1.9%	$93.20 \pm 1.45\%$	4.0
Tier 3: Weather Fusion GBDT (Schema 2)	3.89 ± 1.25	0.2%	$90.72 \pm 1.73\%$	4.0
Tier 4: Physics-Informed Nuclear GBDT (Schema 3)	$\mathbf{2.71 \pm 1.39}$	**0.0%**	$\mathbf{94.00 \pm 0.93\%}$	**4.4**
Tier 4: Physics-Informed Nuclear (90% target)	$\mathbf{2.63 \pm 1.49}$	**0.0%**	$\mathbf{93.82 \pm 1.24\%}$	**4.5**
Tier 3: Multi-Layer Perceptron	60.75 ± 22.86	3.8%	$90.30 \pm 2.75\%$	3.4

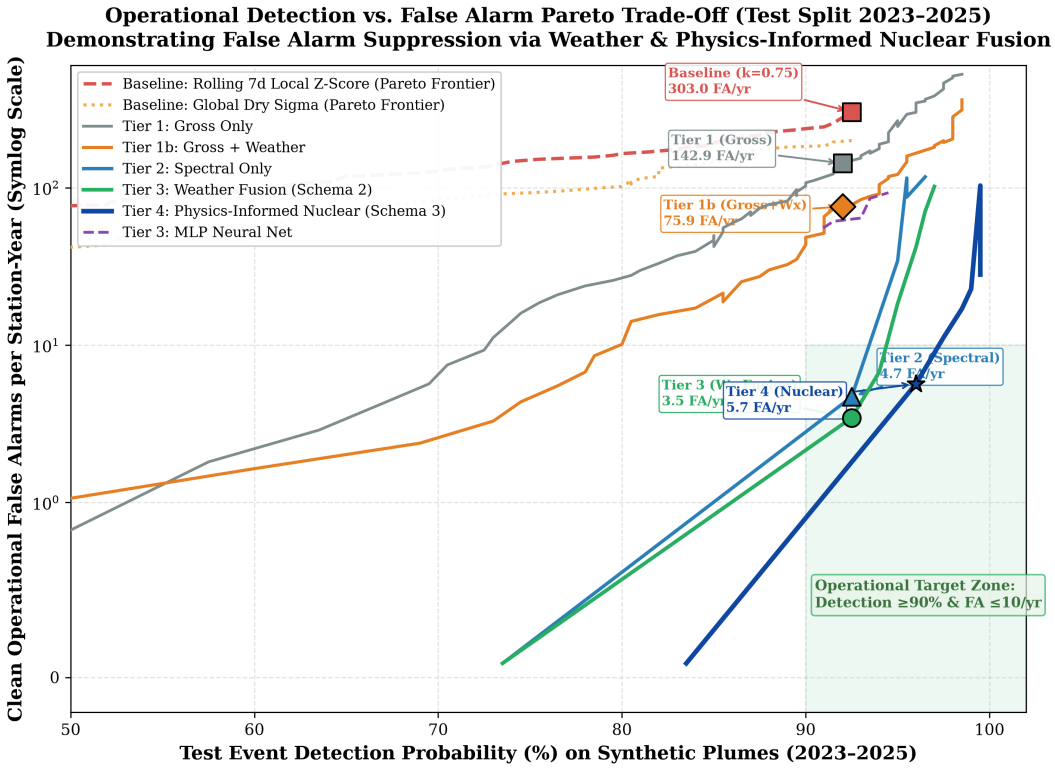


Figure 2: Receiver Operating Characteristic (ROC) matched detection curves comparing baseline rules against machine learning tiers on the test split.

3.8 The Rain-Masked Baseline Control Experiment

To rigorously answer whether complex machine learning is necessary or if a simple heuristic rule mut-

ing alarms during rainfall suffices, an explicit control experiment was developed (src/evaluate_rain_masked_baseline.py). The results at matched detection targets are summarized in Table 3.

Table 3: Rain-Masked Baseline Control Experiment at Matched Detection Targets

Baseline / Mask Rule	**Lowest FA/yr at $\geq 87.12\%$ Det.**	**Lowest FA/yr at $\geq 91.88\%$ Det.**	**Max Det. (%)**
Unmasked ($H =$ none)	215.48 ($k = 1.40$)	303.03 ($k = 0.75$)	92.50%
Rain Mask: Rain only ($H = 0$h)	197.47 ($k = 1.35$)	285.77 ($k = 0.75$)	92.50%
Rain Mask: Rain + 3h ($H = 3$h)	179.80 ($k = 1.30$)	251.65 ($k = 0.75$)	92.50%
Rain Mask: Rain + 6h ($H = 6$h)	181.77 ($k = 1.25$)	245.98 ($k = 0.75$)	92.50%
Rain Mask: Rain + 12h ($H = 12$h)	177.49 ($k = 1.25$)	238.66 ($k = 0.75$)	92.00%
Tier 3 ML (Weather Fusion, Schema 2)	3.89	3.89	96.50%
Tier 4 ML (Nuclear Physics, Schema 3)	**2.63**	**2.71**	**98.00%**

The control experiment demonstrates that simply suppressing alarms during rain fails. To reach $\geq 91.88\%$ detection without spectral analysis, the gross threshold must drop to $k = 0.75\sigma$, causing 238.66 false alarms/year from baseline noise. Tier 3 machine learning achieves a 97.6% false alarm reduction over the best rain-mask heuristic, confirming the scientific necessity of multi-channel spectral and weather fusion.

REFERENCES

Abdelhakim, A., 2025. Anomaly detection in gamma-ray radiation spectra using artificial neural network and ant colony optimization. *Journal of Environmental Radioactivity* 283, 107790. https://doi.org/10.1016/j.jenvrad.2025.107790

Alamaniotis, M., 2024. Mode-driven explainable artificial intelligence approach for estimating background radiation spectrum in a measurement applicable to nuclear security. *Annals of Nuclear Energy* 209, 110849. https://doi.org/10.1016/j.anucene.2024.110849

Ayoub, A., Wainwright, H.M., Sansavini, G., 2024. Machine learning-enabled weather forecasting for real-time radioactive transport and contamination prediction. *Progress in Nuclear Energy* 172, 105255. https://doi.org/10.1016/j.pnucene.2024.105255

Ayoub, A., Wainwright, H.M., Wang, L., Sansavini, G., 2024. An enhanced Fourier neural operator surrogate for radioactive plume transport forecasting. *Stochastic Environmental Research and Risk Assessment* 38(8), 3173–3189. https://doi.org/10.1007/s00477-024-02738-8

Bandstra, M.S., Curtis, J.C., Ghawaly, J.M., Joshi, T.H., Bilton, K.J., 2023. Explaining machine-learning models for gamma-ray detection and identification. *PLOS ONE* 18(6), e0286829. https://doi.org/10.1371/journal.pone.0286829

Bilton, K.J., Joshi, T.H., Bandstra, M.S., Curtis, J.C., Quiter, B.J., Cooper, R.J., Vetter, K., 2019. Non-negative matrix factorization of gamma-ray spectra for background modeling, detection, and source identification. *IEEE Transactions on Nuclear Science* 66(5), 827–837. https://doi.org/10.1109/TNS.2019.2907267

Breitkreutz, H., Mayr, J., Bleher, M., Seifert, S., Stöhlker, U., 2023. Identification and quantification of anomalies in environmental gamma dose rate time series using artificial intelligence. *Journal of Environmental Radioactivity* 259–260, 107082. https://doi.org/10.1016/j.jenvrad.2022.107082

Buckton, C.J., Wyngaardt, S.M., Ngxande, M., 2025. Radioisotope compositional analysis using Monte Carlo γ-ray simulations and regression neural network. *Applied Radiation and Isotopes* 220, 111746. https://doi.org/10.1016/j.apradiso.2025.111746

Burnett, J.L., Croudace, I.W., Warwick, P.E., 2010. Short-lived variations in the background gamma-radiation dose. *Journal of Radiological Protection* 30(3), 525–533. https://doi.org/10.1088/0952-4746/30/3/007

Cetnar, J., 2006. General solution of Bateman equations for nuclear transmutations. *Annals of Nuclear Energy* 33(7), 640–645. https://doi.org/10.1016/j.anucene.2006.02.004

Currie, L.A., 1968. Limits for qualitative detection and quantitative determination. Application to radiochemistry. *Analytical Chemistry* 40(3), 586–593. https://doi.org/10.1021/ac60259a007

Detwiler, R.S., Pfund, D.M., Myjak, M.J., Kulisek, J.A., Seifert, C.E., 2015. Spectral anomaly methods for aerial detection using KUT nuisance rejection. *Nuclear Instruments and Methods in Physics Research Section A* 784, 339–345. https://doi.org/10.1016/j.nima.2015.01.040

Fjeldsted, A., Morrow, T., Scott, C., Hagen, A., 2024. A novel methodology for gamma-ray spectra dataset procurement over varying standoff distances and source activities. *Nuclear Instruments and Methods in Physics Research Section A* 1064, 169681. https://doi.org/10.1016/j.nima.2024.169681

Frankemölle, J.P.K.W., Camps, J., De Meutter, P., Meyers, J., 2025. A Bayesian method for predicting background radiation at environmental monitoring stations in local-scale networks. *Geoscientific Model Development* 18(6), 1989–2003. https://doi.org/10.5194/gmd-18-1989-2025

Hsieh, T.C., Li, C.H., Li, S.H., 2026. A minute-level analysis between environmental gamma dose rate and precipitation. *IEEE Access* 13, 3649414. https://doi.org/10.1109/ACCESS.2025.3649414

ICRP, 1996. Conversion Coefficients for Use in Radiological Protection against External Radiation (ICRP Publication 74). *Annals of the ICRP* 26(3–4), 5–19. https://doi.org/10.1016/S0146-6453(96)90003-2

Karafasoulis, K., Kyriakis, A., 2026. Anomaly detection in gamma-ray spectra using autoencoders with small form factor CZT detectors. *Applied Radiation and Isotopes* 230, 112409. https://doi.org/10.1016/j.apradiso.2025.112409

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems (NeurIPS 2017)* 30, 3146–3154. https://proceedings.neurips.cc/paper_files/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf

Knoll, G.F., 2010. Radiation Detection and Measurement, fourth ed.. *John Wiley & Sons, Hoboken, NJ*. https://www.wiley.com/en-us/Radiation+Detection+and+Measurement%2C+4th+Edition-p-9780470131480

Lalor, P., Adams, H., Hagen, A., 2026. Sim-to-real supervised domain adaptation for radioisotope identification. *Nuclear Instruments and Methods in Physics Research Section A* 1083, 171159. https://doi.org/10.1016/j.nima.2025.171159

Lye, A., Ong, Z.Y., Xiao, N., Patelli, E., 2025. Physics-enhanced machine learning for probabilistic risk assessment in nuclear safety: An overview, recent developments, and perspectives. *Annals of Nuclear Energy* 218, 111562. https://doi.org/10.1016/j.anucene.2025.111562

Masson, O., Baeza, A., Bieringer, J., Brudecki, K., et al., 2011. Tracking of airborne radionuclides from the damaged Fukushima Dai-Ichi nuclear reactors by European networks. *Environmental Science & Technology* 45(18), 7670–7677. https://doi.org/10.1021/es2017158

Melintescu, A., Chambers, S.D., Crawford, J., Williams, A.G., Zorila, B., Galeriu, D., 2018. Radon-222 related influence on ambient gamma dose. *Journal of Environmental Radioactivity* 189, 67–78. https://doi.org/10.1016/j.jenvrad.2018.03.012

Meng, L.J., Ramsden, D., 2000. An inter-comparison of three spectral-deconvolution algorithms for gamma-ray spectroscopy. *IEEE Transactions on Nuclear Science* 47(4), 1329–1336. https://doi.org/10.1109/23.872973

Mercier, J.F., Tracy, B.L., d'Amours, R., Chagnon, F., Hoffman, I., Ungar, R.K., Korpach, E.P., 2009. Increased environmental gamma-ray dose rate during precipitation: a strong correlation with contributing air mass. *Journal of Environmental Radioactivity* 100(7), 527–533. https://doi.org/10.1016/j.jenvrad.2009.03.002

Minato, S., 1980. Analysis of time variations in natural background gamma radiation flux density. *Journal of Nuclear Science and Technology* 17(6), 461–469. https://doi.org/10.1080/18811248.1980.9732610

Moszyński, M., Nassalski, A., Syntfeld-Każuch, A., Szczęśniak, T., Czarnacki, W., Wolski, D., Pausch, G., Stein, J., 2006. Temperature dependences of $LaBr_3$(Ce), $LaCl_3$(Ce) and NaI(Tl) scintillators. *Nuclear Instruments and Methods in Physics Research Section A* 568(2), 739–751. https://doi.org/10.1016/j.nima.2006.06.039

Paatero, J., 2000. Wet deposition of radon-222 progeny in northern Finland measured with an automatic precipitation gamma analyser. *Radiation Protection Dosimetry* 87(4), 273–280. https://doi.org/10.1093/oxfordjournals.rpd.a033008

Sangiorgi, M., Hernández-Ceballos, M.A., Jackson, K., Cinelli, G., Bogucarskis, K., De Felice, L., Patrascu, A., De Cort, M., 2020. The European Radiological Data Exchange Platform (EURDEP): 25 years of monitoring data exchange. *Earth System Science Data* 12(1), 109–118. https://doi.org/10.5194/essd-12-109-2020

Smith, A., Lott, N., Vose, R., 2011. The Integrated Surface Database: Recent developments and partnerships. *Bulletin of the American Meteorological Society* 92(6), 704–708. https://doi.org/10.1175/2011BAMS3015.1

Sportisse, B., 2007. A review of parameterizations for modelling dry deposition and scavenging of radionuclides. *Atmospheric Environment* 41(13), 2683–2698. https://doi.org/10.1016/j.atmosenv.2006.11.057

Stohl, A., Seibert, P., Wotawa, G., Arnold, D., Burkhart, J.F., Eckhardt, S., Tapia, C., Vargas, A., Yasunari, T.J., 2012. Xenon-133 and caesium-137 releases into the atmosphere from the Fukushima Dai-ichi nuclear power plant: determination of the source term, atmospheric dispersion, and deposition. *Atmospheric Chemistry and Physics* 12(5), 2313–2343. https://doi.org/10.5194/acp-12-2313-2012

Sun, J., Niu, D., Liang, J., Li, L., 2024. Rapid nuclide identification algorithm based on self-attention mechanism neural network. *Annals of Nuclear Energy* 207, 110708. https://doi.org/10.1016/j.anucene.2024.110708

Takeuchi, N., Katase, A., 1982. Rainout-washout model for variation of environmental gamma-ray intensity by precipitation. *Journal of Nuclear Science and Technology* 19(5), 393–409. https://doi.org/10.1080/18811248.1982.9734160

Wen, S., Zhou, R., 2024. Temperature peak-drift correction method for NaI(Tl) detectors using the background peak. *Sensors* 24(8), 2621. https://doi.org/10.3390/s24082621